

Derek Stone

San Ramon, CA
contact.stonederek@gmail.com
www.linkedin.com/in/derekstone0

- I'm an Electrical Engineering student at Rose-Hulman Institute of Technology interested in exploring the intersection of optics and hardware.
- I'm passionate about and experienced in exploring new technologies.
- I want to work with companies who are empowered to try/do things that nobody else will.

Skills:

Programming: Python, Java, TensorFlow, PyTorch, GitHub

CAD & Engineering Software: SOLIDWORKS, Onshape

Technical: Signal Processing, Statistical Modeling, Computational Modeling & Simulation

Work Experience:

University of California, Davis – Davis, CA

Research and Development Intern

July 2025 – August 2025

Electrical & Computer Engineering Department

- Co-developed a proof-of-concept Retrieval-Augmented Generation chatbot for UC Davis academic advising using LangChain, vector search, document retrieval, and large language model response generation.
- Engineered a multi-stage query and output validation system to mitigate prompt injection and jailbreak attacks, outperforming GPT-4o on a team-designed adversarial test suite.

Research Intern

April 2025 – June 2026

Broadband Radio Access Technologies Research Lab

- Developed an acoustic drone detection pipeline in Python using TensorFlow and PyTorch, converting raw audio into time-frequency representations for convolutional neural network classification of drone and background noise.
- Improved model robustness through experiments with class balancing, normalization, data augmentation, regularization, learning-rate tuning, architecture changes, and transfer learning.
- Evaluated generalization on unseen recordings using sliding-window inference, confusion matrices, false-alarm rates, and model-interpretability tools to identify dataset bias and distribution shift.

Project Experience:

2D Lightning Simulation

April 2025

- Developed a 2D lightning propagation simulation in NetLogo using an electric potential field, stochastic path selection, branching logic, and procedurally generated terrain.
- Translated physical concepts into computational rules and implemented adjustable parameters to model variation in lightning paths and strike behavior.

Education:

Bachelor of Science, Electrical Engineering

Minor in Optical Engineering

May 2030

Rose-Hulman Institute of Technology – Terre Haute, IN

Relevant Coursework: DC Circuits, Differential Equations II, Physics III, Intro to Signal Proc.

High School Diploma

May 2026

Dougherty Valley High School – San Ramon, CA

GPA: 4.23

Relevant Coursework: AP Calculus BC, AP Computer Science A, AP Physics C: Mechanics

Activities/Personal Interests:

Rose Propulsion Laboratory, Church AV Technician, Peer Tutor, First Degree Kajukenbo Black Belt, Weightlifting, Climbing, Digital & Traditional Art